Weekly Self-Study Tasks to Complete

Containerization with Docker for Leaders

This document covers all of Week 7 of Phase 3 in the 4-hour Leaders model: 2 hours of instructor-led session, 1 hour of self-study video, and 1 hour of hands-on lab. The Core Resources table below fits inside the 1-hour self-study media budget. The Practice Activities section points to the Evergreen container lab, where you will build and run the API as a Docker container. Additional Content at the end is optional and sits outside the 4-hour budget.

Core Resources (~60 min)

Watch the specific resources called out below — not an entire course. Keep a terminal open so you can run the docker commands the videos show.

Topic	Material Description	Est. Time	Link / Reference
Docker Big Picture	Docker for Data Engineers (Janani Ravi, LinkedIn Learning, released Oct 18, 2024; 2h 30m course). Watch "Introducing Docker" and "Working with Docker" (~25 min) — general Docker fundamentals: what containers are, how they differ from VMs, and how images and containers work.	~25 min	linkedin.com/learning — Docker for Data Engineers
Images & Dockerfiles	Docker "Get Started" — "What is an image?" and "Build and push your first image" (official docs). Read and run the build/run steps.	~20 min	docs.docker.com — Get Started What is an image? | Docker Docs
Containers in Practice	Docker "Get Started" workshop (official guide) — read and skim how a real app is packaged and run as a container.	~15 min	docs.docker.com — Workshop

Practice Activities (1 hour)

Complete this week's hands-on lab: "Evergreen Insurance Quote API — Containerizing with Docker." You will write a Dockerfile, build an image, and run the API inside a container. The whole exercise takes about one hour.
In the lab you will:

- Open the Labfiles/evergreen-quote-docker folder and confirm Docker with docker --version.
- Write the Dockerfile: choose a small Node base image, copy and install, copy source, set the start command.
- Order the instructions so dependency install is cached separately from source (layer caching).
- Build the image with docker build -t evergreen-quote-api .
- Run it with docker run --rm -p 3000:3000 evergreen-quote-api and curl the API.
- Discuss why the image runs the same anywhere — the foundation for orchestration.

The lab instructions (with "Why this matters" callouts throughout) are in the lab environment. Budget one hour and work at your own pace.

Additional Content (Optional)

The following resources are optional and recommended for students who want to go deeper beyond the core curriculum. They are outside the 4-hour weekly budget. Next week orchestrates this container on Kubernetes/EKS, so focus this week on images, Dockerfiles, and running containers.

Topic	Material Description	Link / Reference
Docker	The Docker documentation — the authoritative guide to images, Dockerfiles, volumes, networking, and Compose (Reference).	docs.docker.com
Dockerfiles	Best practices for writing Dockerfiles — official Docker guidance on layers,	docs.docker.com — Dockerfile best practices

Topic	Material Description	Link / Reference
	caching, and small images (Reference).	
Compose	Docker Compose overview — running multi-container apps (e.g., API + database) from one file (Reference).	docs.docker.com/compose
Security	Docker security — official guidance on minimizing image size and vulnerabilities for production (Reference).	docs.docker.com — security